

UI/UX DESIGN & FUNCTIONAL SPECIFICATION

Business Process Copilot

Operational Data Source Recommendation for Medallion Architecture

Version 1.0

July 8, 2026

Prepared for: Engineering, Data Platform & Product Design

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Table of Contents

1. Introduction & Purpose

The Business Process Copilot is an AI-assisted application that helps data engineers, analysts, and business stakeholders identify, evaluate, and map the right operational data sources into a Medallion (Bronze / Silver / Gold) architecture. Instead of manually researching source systems, schemas, and lineage, users describe a business process in natural language and the copilot recommends candidate sources, assesses data quality and governance risk, and proposes a layer-by-layer ingestion path.

This document defines the information architecture, page inventory, and page-level UI/UX specifications needed to design and build the application. It is intended for product designers, frontend/backend engineers, and reviewers who need a shared reference before implementation begins.

1.1 Problem Statement

- Operational data source selection today is manual, tribal-knowledge-driven, and inconsistent across teams.
- Analysts frequently duplicate discovery effort or choose sources with poor lineage, freshness, or governance posture.
- There is no single place to translate a business process into a validated Bronze → Silver → Gold ingestion plan.

1.2 Goals & Success Criteria

- Reduce time-to-source-selection from days to minutes via a conversational copilot.
- Standardize recommendations against governance, PII, freshness, and lineage rules.
- Provide auditable, exportable recommendation artifacts for engineering handoff.
- Give admins visibility and control over the source catalog, users, and copilot configuration.
- Let any user ask free-form questions about the metadata catalog itself, independent of running a full recommendation.

2. Users & Roles

The application supports three primary roles, enforced consistently across every page:

Role	Description	Representative permissions
Viewer	Business stakeholder who reviews recommendations	Read-only access to results, history, and catalog
Analyst	Data analyst / engineer running recommendations	Create & run recommendations, edit drafts, export specs
Admin	Platform owner managing governance & access	Manage catalog, users, policies, and copilot configuration

3. Information Architecture

The diagram below shows the full page inventory and primary navigation paths, from authentication through the recommendation wizard and into the administrative surfaces.

Information Architecture — Business Process Copilot for Data Source Recommendation

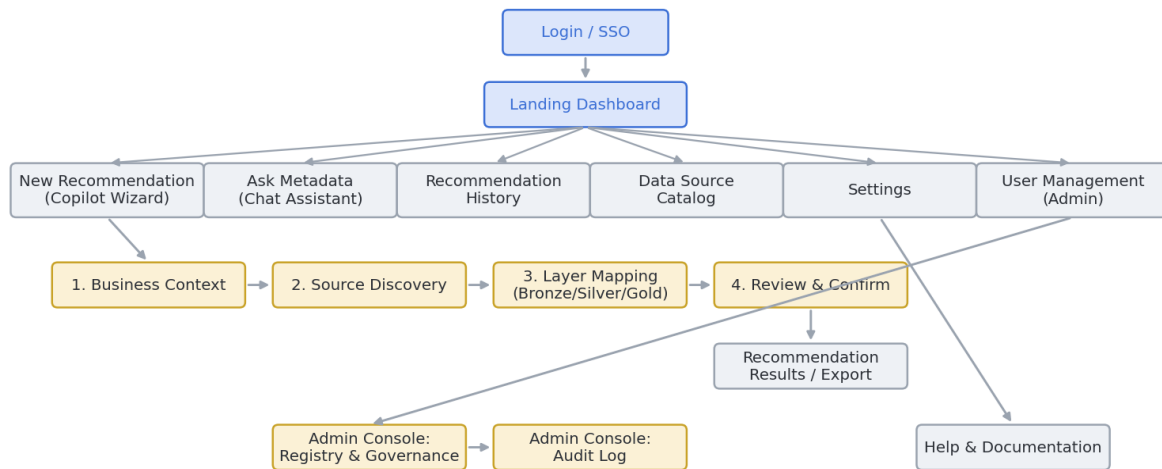


Figure: Site map and primary navigation flow

3.1 Global Navigation Pattern

- Persistent left sidebar for authenticated pages: Home, New Recommendation, Ask Metadata, Recommendation History, Data Source Catalog, Settings, User Management, Admin Console (role-gated).
- Persistent top header: app branding, global search, notifications, and account menu on every authenticated page.
- Sidebar items are filtered by role — Viewers do not see User Management or Admin Console; Analysts do not see Admin Console.
- Breadcrumbs and a step indicator are used inside the multi-step Recommendation Wizard only.

4. Page-by-Page Specifications

4.1. Login Page

Entry point for authentication. Supports both direct email/password login and enterprise SSO (Azure AD / Okta), consistent with organizational identity requirements.

Primary users: All roles

Key UI elements

- App logo and product name
- Email and password fields with inline validation
- “Remember me” checkbox and “Forgot password” link

- Primary “Sign In” button
- Secondary “Sign in with SSO” button
- Access-request messaging for new users

User flow

- User lands on Login page (direct link or session expiry redirect).
- User authenticates via email/password or SSO.
- On success, role and domain permissions are resolved and the user is routed to the Landing Dashboard.
- On failure, inline error messaging is shown; no distinction between “wrong email” vs “wrong password” is surfaced, for security.

Wireframe

1. Login Page

The wireframe shows a login page for 'Data Source Copilot'. At the top, there is a header with a diamond icon and the text 'Data Source Copilot'. Below this is a central card titled 'Welcome back'. Inside the card, there are two input fields: 'Work email' and 'Password'. Below the 'Password' field, there is a checkbox labeled 'Remember me' and a link labeled 'Forgot password?'. A prominent blue 'Sign In' button is positioned below these fields. Underneath the button is a link that says 'Sign in with SSO (Azure AD / Okta)'. At the bottom of the card, there is a line of text: 'New team member? Request access from your admin'. Below the card, at the very bottom of the page, is a small footer text: 'Auth: SSO / OAuth2 + email-password • Roles assigned post-login (Viewer / Analyst / Admin)'.

Figure: Login Page — low-fidelity wireframe (illustrative layout, not final visual design)

4.2. Landing / Home Dashboard

Role-aware home page summarizing recent activity, key metrics, and a fast path into starting a new recommendation.

Primary users: Viewer, Analyst, Admin (content varies by role)

Key UI elements

- Summary metric cards (active requests, sources recommended, pipelines mapped, average confidence)
- Recent recommendation runs list with status
- Copilot suggestions / proactive insights panel
- Primary call-to-action: “Start New Recommendation”
- Global search bar in header

User flow

- User lands here immediately after login.
- User scans recent runs and suggestions, or clicks a run to view details.
- User clicks “Start New Recommendation” to enter the wizard, or uses the sidebar to navigate elsewhere.

Wireframe

2. Landing / Home Dashboard

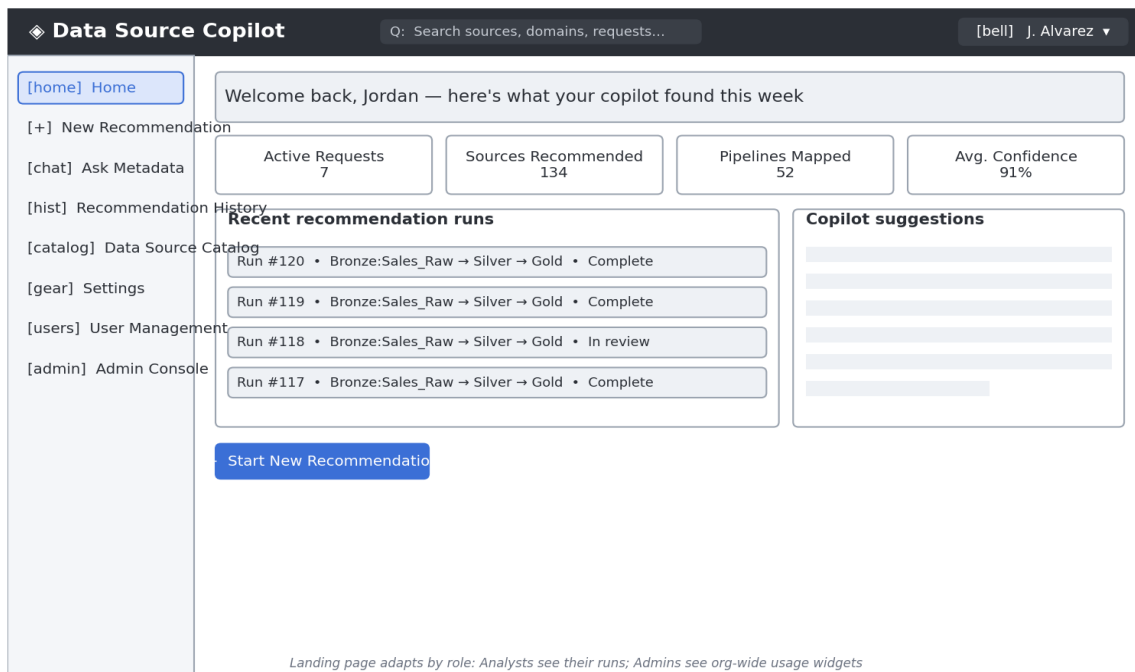


Figure: Landing / Home Dashboard — low-fidelity wireframe (illustrative layout, not final visual design)

4.3. Ask Metadata — Conversational Catalog Assistant

A standalone, always-available chatbot where users ask free-form questions about the data estate itself — schemas, columns, ownership, lineage, freshness, and governance flags — without needing to run a full recommendation. Every answer is grounded in the live metadata catalog and cites the specific tables/columns it draws from.

Primary users: Viewer, Analyst, Admin

Key UI elements

- Persistent chat panel with conversation history (user and copilot turns)
- Suggested-prompt chips for common questions (e.g., “Which tables have customer PII?”, “Lineage for fct_orders”)
- Free-text input for ad-hoc natural-language questions
- “Referenced metadata” side panel listing the exact tables/columns cited in the current answer, each with a direct “Open in Catalog” link
- Optional scope selector to constrain answers to a domain, system, or medallion layer

User flow

- User opens Ask Metadata from the persistent sidebar (available from any page).
- User types a question or selects a suggested prompt, e.g., “Which Silver-layer tables contain customer PII?”
- Copilot answers in natural language and populates the Referenced Metadata panel with the specific entities the answer relies on.
- User asks follow-up questions in the same thread (e.g., “Who owns this table?”); context carries across turns.
- User clicks “Open in Catalog” on any referenced entity to jump to its full detail page in the Data Source Catalog.
- Conversations are not required to start a recommendation, but users can pivot from a chat answer into “Start New Recommendation” with that context pre-filled.

Wireframe

11. Ask Metadata — Conversational Catalog Assistant

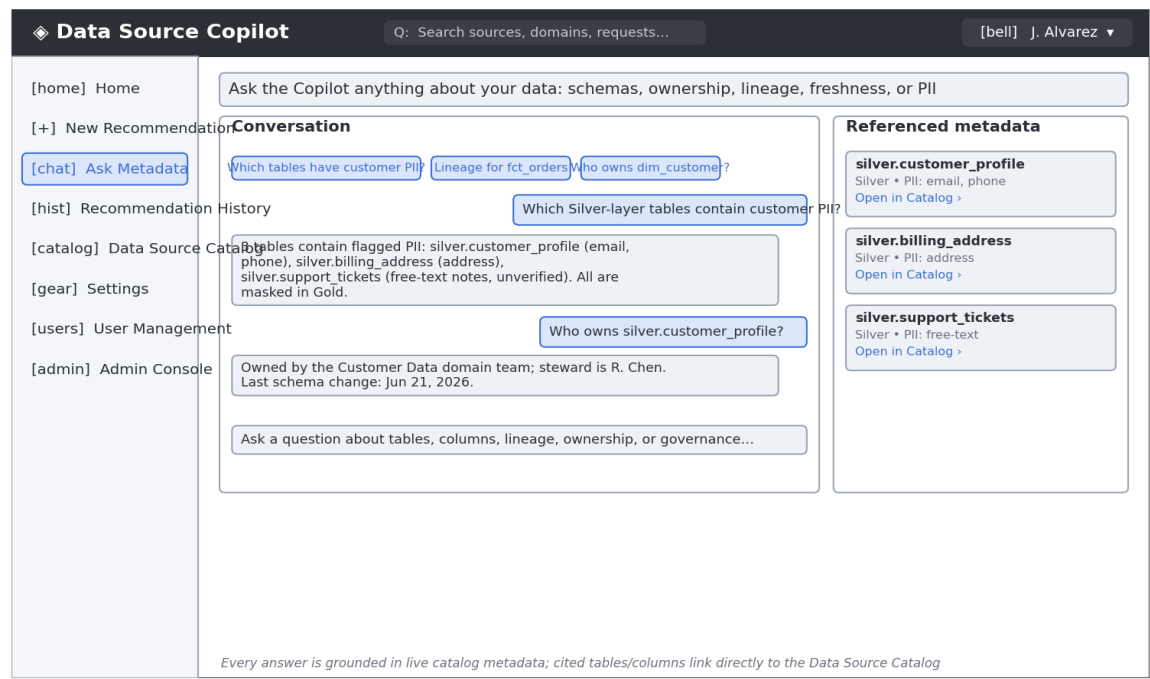


Figure: Ask Metadata — Conversational Catalog Assistant — low-fidelity wireframe (illustrative layout, not final visual design)

4.4. New Recommendation — Copilot Intake Wizard (Step 1–2)

Conversational intake where the user describes the business process in natural language; the copilot asks clarifying questions and surfaces candidate operational sources in real time.

Primary users: Analyst, Admin

Key UI elements

- 4-step progress indicator (Business Context → Source Discovery → Layer Mapping → Review & Confirm)
- Chat-style conversation panel with copilot and user turns
- Free-text input with send action
- Live-updating candidate source cards with confidence scores
- Back / Continue navigation

User flow

- User describes the business process and constraints (e.g., domain, refresh cadence, target consumers).
- Copilot asks clarifying questions as needed and queries the metadata catalog in the background.
- Candidate sources populate on the right with a confidence score and layer classification.
- User refines via chat (e.g., “exclude legacy systems”) and the candidate list updates live.
- User proceeds to Layer Mapping once satisfied with the candidate set.

Wireframe

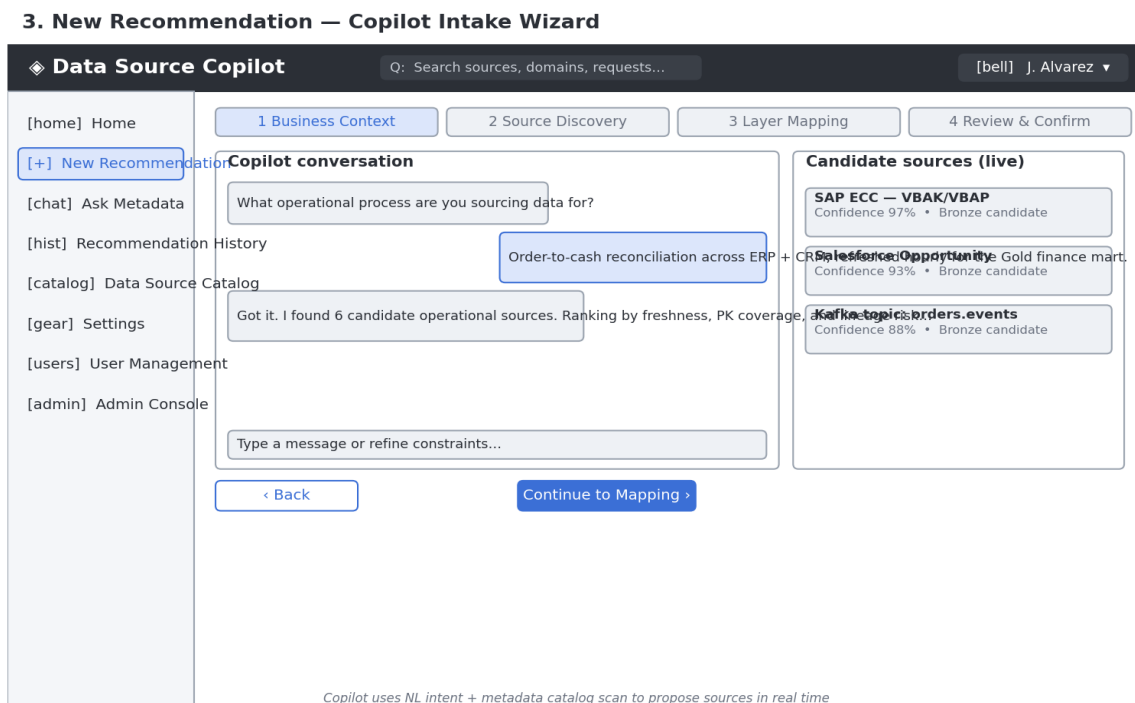


Figure: New Recommendation — Copilot Intake Wizard (Step 1–2) — low-fidelity wireframe (illustrative layout, not final visual design)

4.5. Medallion Layer Mapping View (Step 3)

Visual, drag-and-drop mapping of selected sources into Bronze, Silver, and Gold layers, with live lineage and SLA validation.

Primary users: Analyst, Admin

Key UI elements

- Three-column Kanban-style board: Bronze, Silver, Gold
- Draggable source/table cards between columns
- Inline lineage and dependency arrows between layers
- Validation summary bar (lineage confidence, PK/FK integrity, freshness SLA)
- Back / Continue to Review navigation

User flow

- System pre-populates Bronze with raw candidate sources from Step 2.
- User drags cards into Silver / Gold to define the target conformance and consumption layer.
- Copilot re-validates lineage, key integrity, and SLA feasibility after every change.
- User proceeds to Review & Confirm once validation passes.

Wireframe

4. Medallion Layer Mapping View

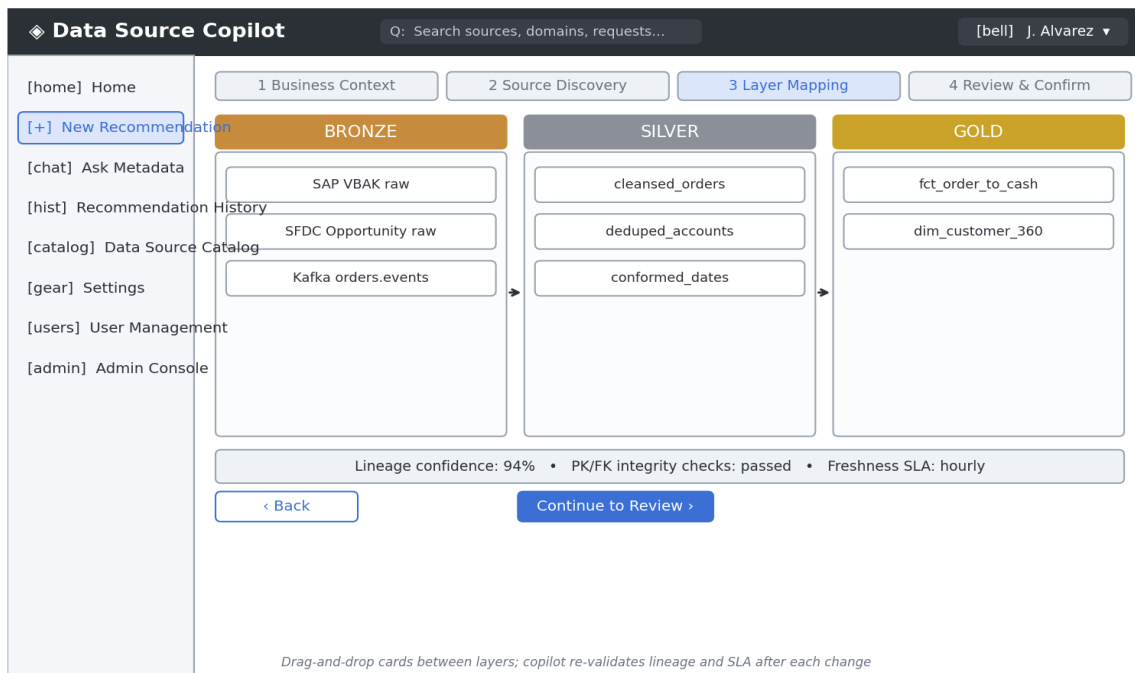


Figure: Medallion Layer Mapping View (Step 3) — low-fidelity wireframe (illustrative layout, not final visual design)

4.6. Review & Recommendation Results (Step 4)

Final summary of the recommended sources, rationale, and governance checks, with actions to approve, export, or hand off to engineering.

Primary users: Analyst, Admin (Viewer: read-only via History)

Key UI elements

- Recommended sources list grouped by layer
- Rationale panel explaining why sources were chosen
- Governance & risk checklist (PII detection, data ownership, retention, cost estimate)
- Primary actions: Approve & Generate Spec, Export as PDF, Send to Data Engineering

User flow

- User reviews the recommended sources and rationale.
- User reviews governance checks; unresolved risk items block approval until acknowledged.
- User approves to generate a formal ingestion spec, exports a PDF summary, or routes it to the data engineering queue.
- Approved recommendations are written to Recommendation History with a version/audit trail.

Wireframe

5. Review & Recommendation Results

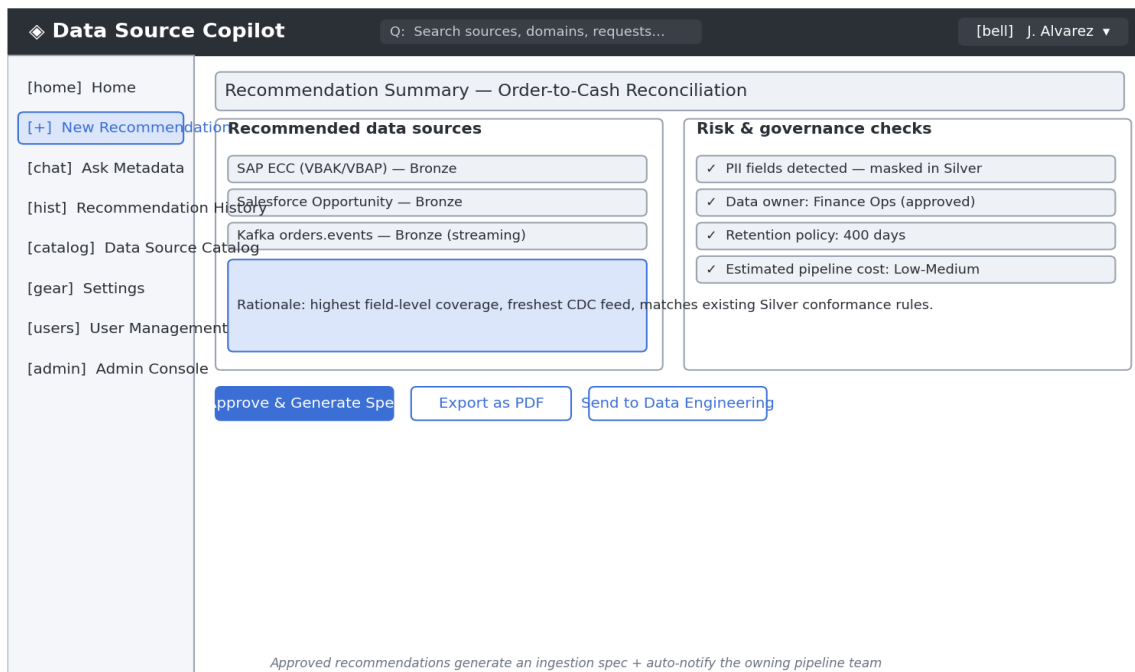


Figure: Review & Recommendation Results (Step 4) — low-fidelity wireframe (illustrative layout, not final visual design)

4.7. Recommendation History

Searchable, filterable log of past and in-progress recommendation requests for auditability and reuse.

Primary users: Viewer (read-only), Analyst, Admin

Key UI elements

- Filter bar (domain, status, date range)
- Sortable table: request name, domain, status, confidence, date
- Row-level click-through to a read-only detail view with audit trail

User flow

- User filters or searches for a past request.
- User clicks a row to open the detail view.
- Viewers can read but not edit; Analysts/Admins can duplicate a past request as a starting point for a new one.

Wireframe

6. Recommendation History

◆ Data Source Copilot

Q: Search sources, domains, requests...

[bell] J. Alvarez ▾

[home] Home

[+] New Recommendation

[chat] Ask Metadata

[hist] Recommendation History

[catalog] Data Source Catalog

[gear] Settings

[users] User Management

[admin] Admin Console

Q: Filter by domain, status, date

REQUEST	DOMAIN	STATUS	CONFIDENCE	DATE
Order-to-Cash Recon	Finance	Approved	94%	Jul 06
Customer 360 Refresh	Sales	In Review	88%	Jul 05
Inventory Aging Report	Supply Chain	Approved	91%	Jul 03
Marketing Attribution	Marketing	Draft	76%	Jul 01
Vendor Risk Scoring	Procurement	Approved	95%	Jun 29

Row click opens the full recommendation detail (read-only) with version/audit trail

Figure: Recommendation History — low-fidelity wireframe (illustrative layout, not final visual design)

12

4.8. Data Source Catalog

Browsable, searchable inventory of known operational and analytical data sources that power the copilot's discovery step.

Primary users: Viewer, Analyst, Admin (edit rights: Admin only)

Key UI elements

- Faceted filters: system type, domain owner, medallion layer, PII flag, freshness SLA
- Search bar for table/system/tag lookup
- Source cards summarizing system, dataset, and layer
- Detail drill-down showing schema, ownership, and lineage (Admin can edit)

User flow

- User browses or searches for a specific system or dataset.
- User applies filters to narrow by domain, layer, or governance flag.
- User opens a source card to view schema and lineage detail.
- Admins can edit catalog metadata directly from this view.

Wireframe

7. Data Source Catalog (Browse & Search)

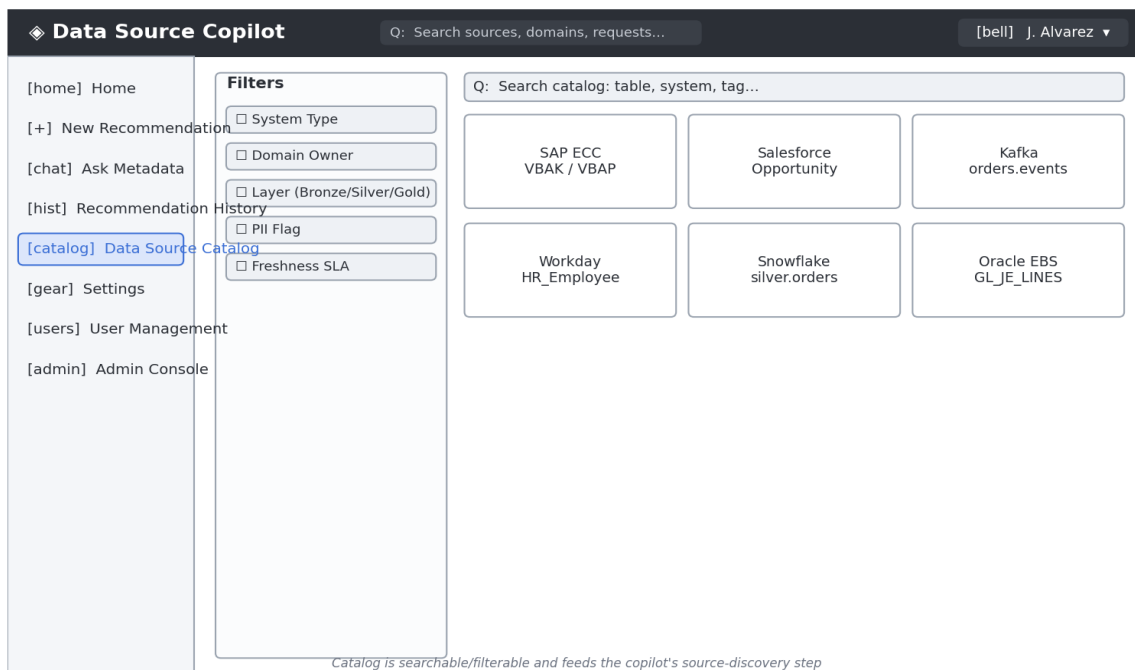


Figure: Data Source Catalog — low-fidelity wireframe (illustrative layout, not final visual design)

4.9. Settings

Personal account, preference, integration, and notification configuration.

Primary users: Viewer, Analyst, Admin

Key UI elements

- Tabbed layout: Profile, Preferences, Integrations, Notifications, API Keys
- Profile fields: name, email, default business domain, default target medallion layer
- Preferences: copilot verbosity, default confidence threshold, units, theme
- Integrations: connected catalogs, BI tools, ticketing systems

User flow

- User opens Settings from the sidebar or account menu.
- User edits a tab and saves changes independently per tab.
- API Keys tab (Analyst/Admin) allows generating and revoking tokens for programmatic access.

Wireframe

8. Settings

Data Source Copilot Q: Search sources, domains, requests... [bell] J. Alvarez ▼

[home] Home [gear] Settings

[+] New Recommendation [chat] Ask Metadata [hist] Recommendation History [catalog] Data Source Catalog [users] User Management [admin] Admin Console

Profile settings

Full name

Work email

Default business domain

Default medallion target layer

Save Changes

Preferences tab controls copilot verbosity, default confidence threshold, units, theme

Figure: Settings — low-fidelity wireframe (illustrative layout, not final visual design)

4.10. User Management (Admin)

Administrative surface for inviting users, assigning roles, and scoping domain access.

Primary users: Admin only

Key UI elements

- Search bar and “Invite User” action
- Table: name, role, domain access, status, last active
- Row-level actions: edit role, edit domain scope, deactivate, resend invite

User flow

- Admin invites a new user by email and assigns an initial role.
- Admin edits role or domain access as responsibilities change.
- Admin deactivates users who no longer need access.

Wireframe

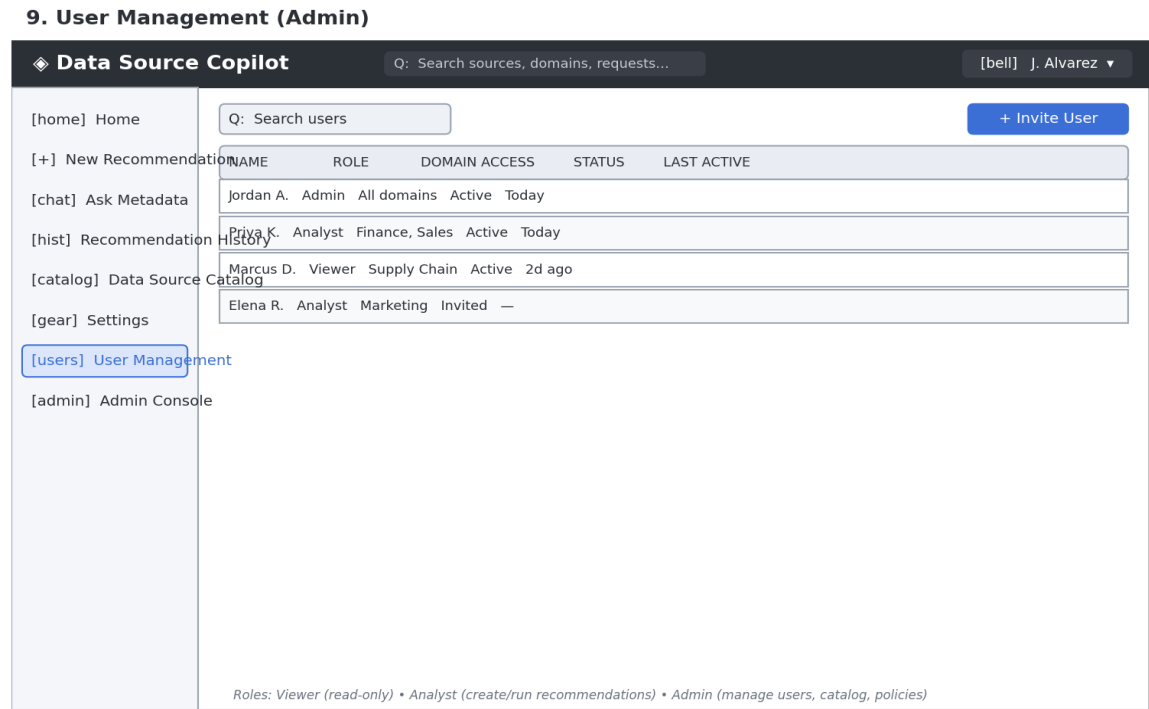


Figure: User Management (Admin) — low-fidelity wireframe (illustrative layout, not final visual design)

4.11. Admin Console — System & Governance

Central control plane for the data source registry, governance policies, copilot/model configuration, audit logging, and usage/cost visibility.

Primary users: Admin only

Key UI elements

- Tabbed layout: Data Source Registry, Governance Policies, Model & Copilot Config, Audit Log, Usage & Cost
- Audit log table: timestamp, user, action, entity
- Governance policy editor (PII handling, retention, approval workflows)
- Copilot configuration (model version, confidence thresholds, allowed data domains)

User flow

- Admin reviews the audit log for compliance or troubleshooting.
- Admin updates governance policies that the copilot enforces during recommendations.
- Admin monitors usage and cost trends for the copilot service.

Wireframe

10. Admin Console — System & Governance

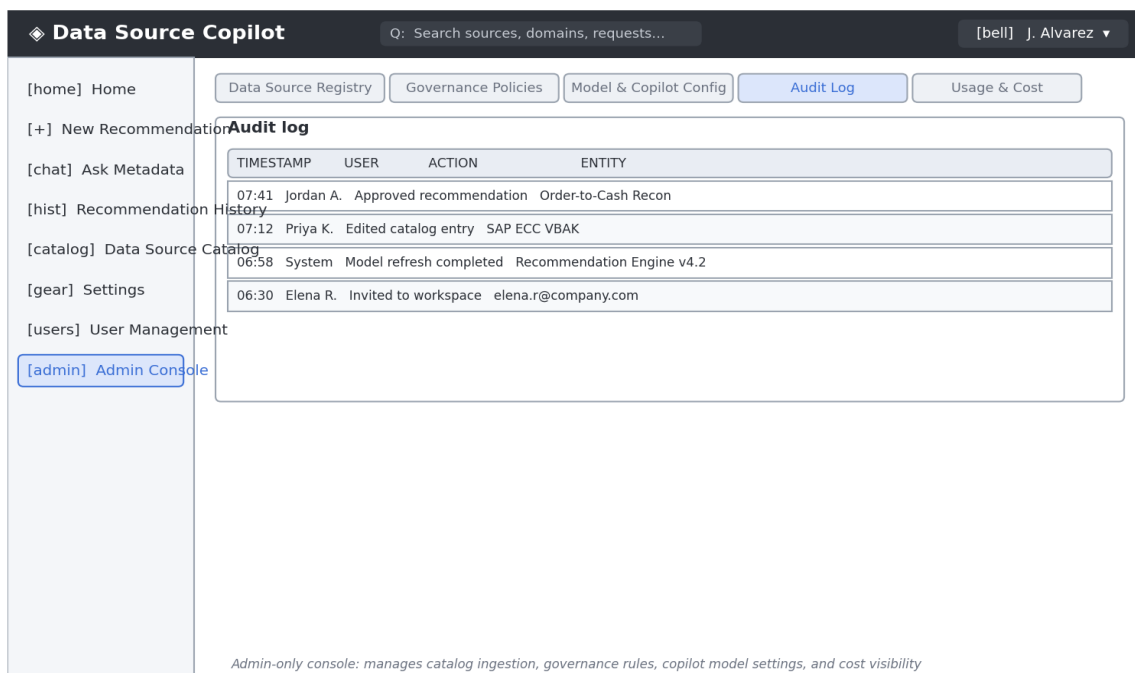


Figure: Admin Console — System & Governance — low-fidelity wireframe (illustrative layout, not final visual design)

5. Cross-Cutting Requirements

5.1 Accessibility

- WCAG 2.1 AA color contrast across all pages, including status and confidence indicators.
- Full keyboard navigation and visible focus states, including within the drag-and-drop layer mapping view (with a non-drag fallback, e.g., “Move to...” menu).
- Screen-reader labels for icon-only controls (search, notifications, filters).

5.2 Performance

- Copilot conversational responses should stream incrementally rather than block on full completion.
- Catalog search and history filtering should return results in under 1 second for typical result sets.

5.3 Security & Governance

- SSO/OAuth2 required for enterprise deployments; session timeout enforced with re-authentication.
- Role-based access control enforced at both the UI (page/element visibility) and API layer.
- PII detected in recommended sources must be flagged before a recommendation can be approved.
- All approvals, edits, and admin actions are written to the immutable audit log.

5.4 Responsiveness

- Primary target is desktop/laptop (data-dense workflows); tablet support for read-only review (History, Results, Catalog).

6. Data & Medallion Architecture Context

The application does not move data itself; it recommends and documents an ingestion path. Recommendations classify sources into the standard medallion layers:

Layer	Definition	Copilot's role
Bronze	Raw operational source data, minimally transformed	Identify and rank candidate operational sources
Silver	Cleansed, conformed, deduplicated data	Propose conformance rules and validate key integrity
Gold	Business-level, consumption-ready datasets	Propose target fact/dimension models for the process

7. Open Questions

- Which metadata catalog(s) (e.g., internal registry, Collibra, Purview) will back live source discovery?
- What is the approval workflow when governance checks fail — hard block vs. override with justification?
- Should recommendation exports integrate directly with ticketing (e.g., Jira) for engineering handoff?